

LEARN PYTHON PROGRAMMING – BEGINNER TO MASTER

Section: 7. String and its Methods

Lecture: 83. Palindrome String - Project

Section 1: Lecture Summary

The lecture explains the concept of palindrome strings and phrases with examples, then guides through a practical Python project to detect and handle palindromes. It starts by defining palindrome words (same forwards and backwards, ignoring case), demonstrates palindrome phrases (ignoring spaces and cases), and then discusses how to create a palindrome from a non-palindrome string by concatenating it with its reverse. The project requires writing a Python program that takes an input string, removes spaces, normalizes case, and checks if it's a palindrome. If it is, the program prints the original string; otherwise, it constructs and prints a palindrome by concatenating the normalized string with its reverse. The lecturer explains the approach step-by-step and demonstrates the code execution.

Section 2: Key Concepts and Explanations

- Palindrome String:

- A string that reads the same forwards and backwards.
- Example: "Madam" → reversed "madam" (case ignored → equal).

- Case Insensitivity:

- Palindrome check ignores letter casing using methods like `.casefold()` to compare strings without case sensitivity.

- Palindrome Phrase:

- A phrase or sentence that is palindrome ignoring spaces and cases.
- Example: "Name now one man" → after removing spaces "namenowoneman" → reversed also same.

- Removing Spaces and Case Normalization:

- Spaces must be removed to check palindrome phrases using string replace method: `s.replace(' ', '')`.`
- Convert to lower case or use `s.casefold()` to manage case differences.

- Making a Palindrome from Non-Palindrome:

- If a string is not a palindrome, create one by concatenating the string with its reversed version.
- Example: For "abcde", palindrome can be "abcde" + "edcba".

- Python Methods Used:

- `s.replace(' ', '')` - remove spaces`
- `s.casefold()` - case insensitive comparison
- `s[::-1]` - slicing to reverse a string

Section 3: Example Code and Use Cases

Take input string

Remove spaces

Get reverse

Check palindrome ignoring case

Create palindrome by concatenation

```
s1 = input("Enter a phrase: ")

s2 = s1.replace(' ', '')

rev = s2[::-1]

if s2.casefold() == rev.casefold():
    print(s1) # Palindrome: print original input
```

```
else:  
    palindrome = s2.casefold() + rev.casefold()  
    print(palindrome) # Print created palindrome
```

Explanation:

- Input a phrase.
- Spaces are removed from the original string.
- Reverse is obtained by slicing.
- The non-space strings are compared in a case-insensitive way using `.casefold()`.
- If palindrome, original string printed; if not, palindrome is created by concatenating the lowercase string and its reverse and printed.

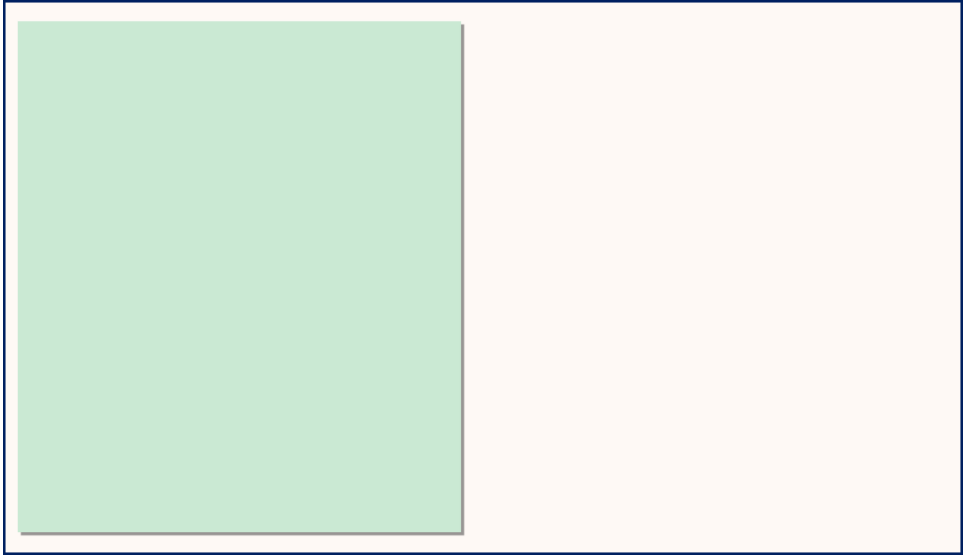
Section 4: Key Takeaways

- A palindrome reads identically forwards and backwards ignoring case.
- Palindromes can be single words or entire phrases/sentences ignoring spaces and cases.
- Use `str.replace(' ', '')` to remove spaces, `str.casefold()` for case-insensitive comparison, and string slicing `[::-1]` to reverse a string in Python.
- Non-palindrome strings can be converted into palindromes by concatenating the string with its reversed form.
- The project exercises string manipulation, conditional logic, and user input handling in Python.

Lecture in Slides

Please Note: This document presents a curated selection of slides from the lecture; others have been excluded for conciseness.





Palindrome String
Project

Palindrome
Project

Palindrome
Project

Palindrome

'Madam'

Palindrome

Project

Palindrome

'Madam'

'madaM'

Palindrome

Project

Palindrome

'madam'

'madam'

Palindrome

Project

Palindrome

'madam'

'madam'

'Malayalam'

Palindrome

Project

Palindrome

'madam'

'madam'

'Malayalam'

'malayalam'

	Palindrome Project Palindrome Phrase
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Palindrome
Project

Palindrome Phrase

'Name now one man'

Palindrome
Project

Palindrome Phrase

'Name now one man'
'Namenowoneman'

Palindrome

Project

Palindrome Phrase

'Name now one man'

'namenowoneman'

Palindrome

Project

Palindrome Phrase

'Name now one man'

'namenowoneman'

'Never odd or even'

Palindrome Project

Palindrome Phrase

'Name now one man'

'namenowoneman'

'Never odd or even'

'neveroddoreven'

Palindrome
Project

Palindrome Phrase

'Name now one man'
'namenowoneman'

'Never odd or even'
'neveroddoreven'

'Was it a car or a cat I saw'

Palindrome
Project

Palindrome Phrase

'Name now one man'
'namenowoneman'

'Never odd or even'
'neveroddoreven'

'Was it a car or a cat I saw'
'wasitacaroracatisaw'

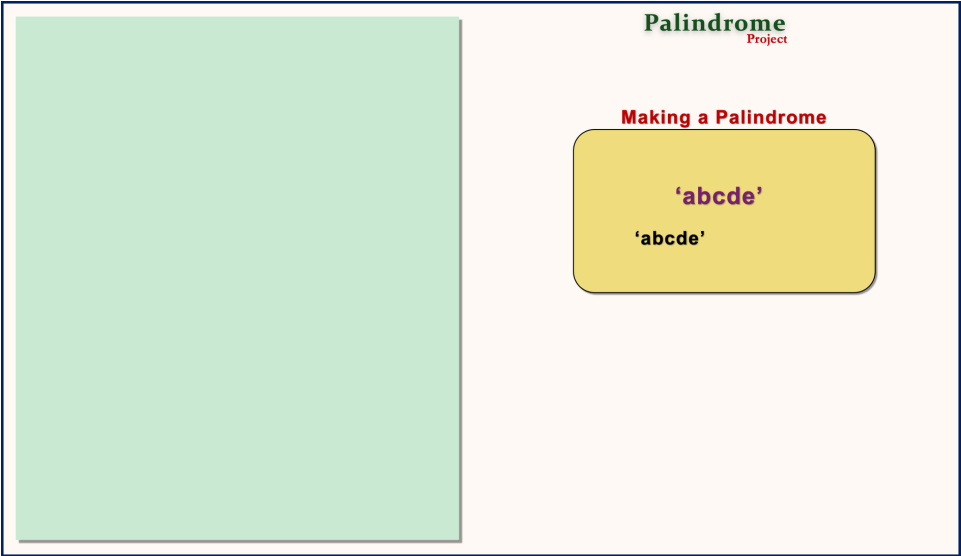
Palindrome
Project

Making a Palindrome

Palindrome
Project

Making a Palindrome

'abcde'



Palindrome

Project

Making a Palindrome

'abcde'

'abcde' + 'edcba'

Palindrome

Project

Making a Palindrome

'abcde'

'abcdeedcba'

Palindrome
Project

Making a Palindrome

'abcde'

'abcdeedcba'

'Python'

Palindrome
Project

Making a Palindrome

'abcde'

'abcdeedcba'

'Python'

'python'

Palindrome Project

Making a Palindrome

'abcde'

'abcdeedcba'

'Python'

'python' + 'ohtyp'

Palindrome
Project

Making a Palindrome

'abcde'

'abcdeedcba'

'Python'

'pythonohtyp'

Palindrome
Project

Project Palindrome

Palindrome

Project

Project Palindrome

'Race car'

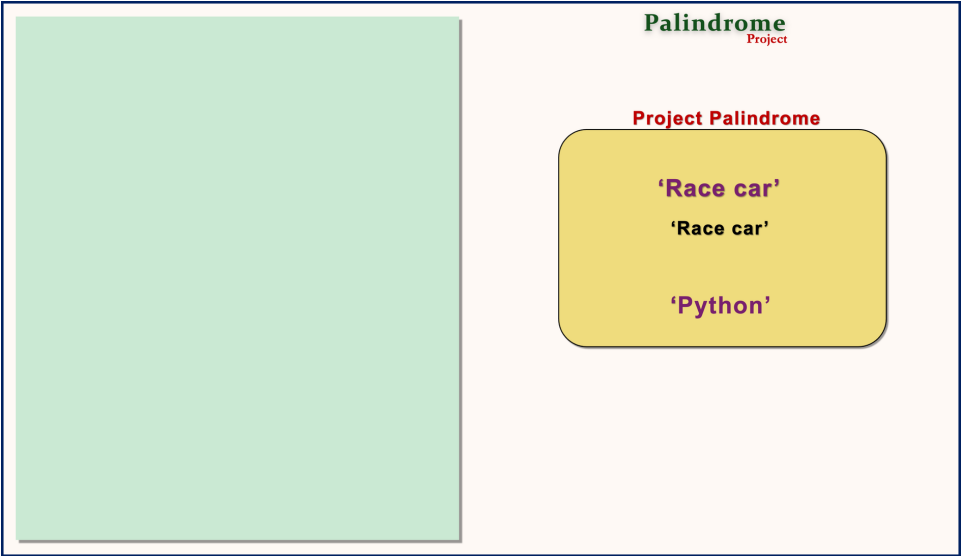
Palindrome

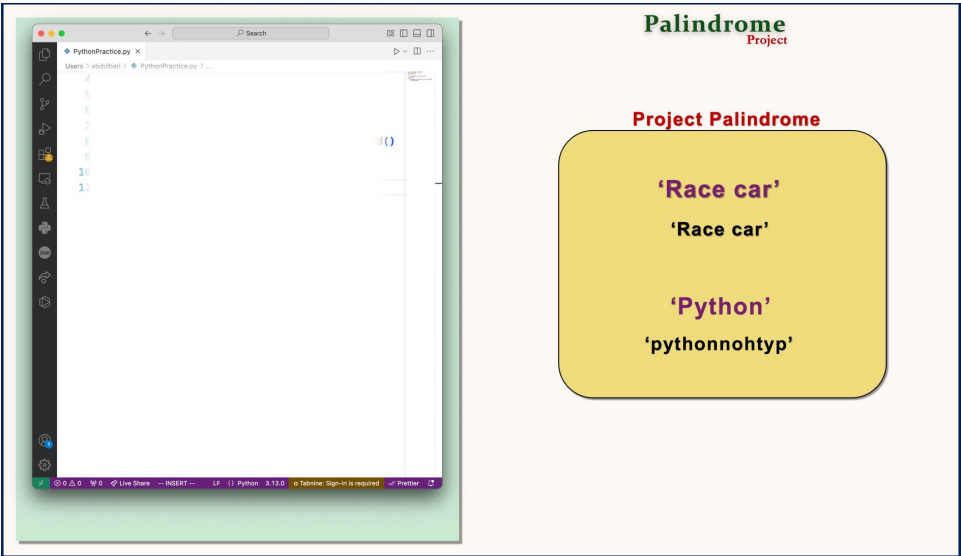
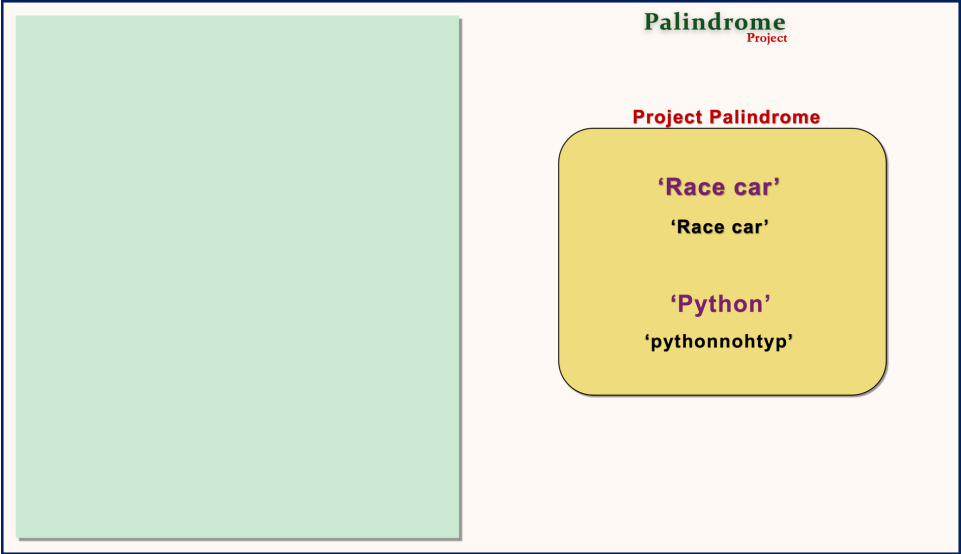
Project

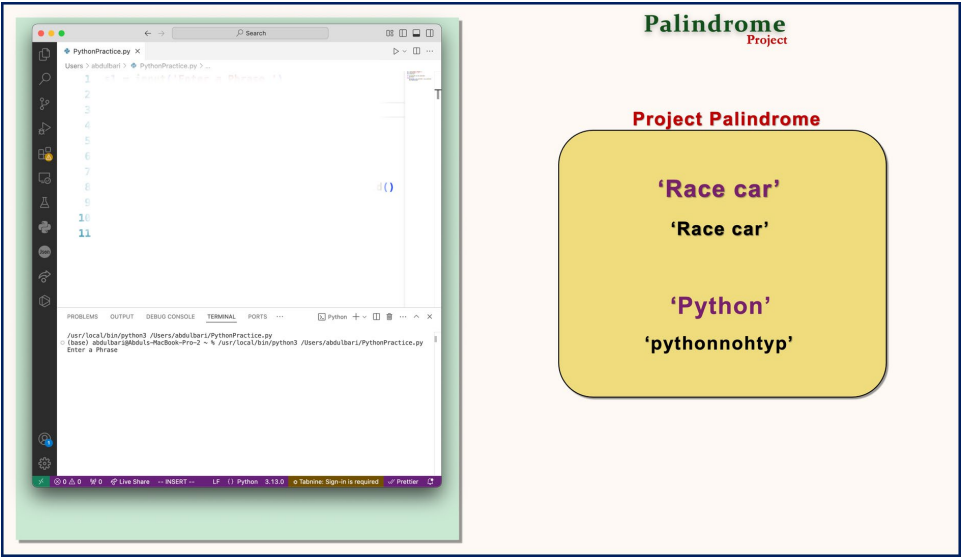
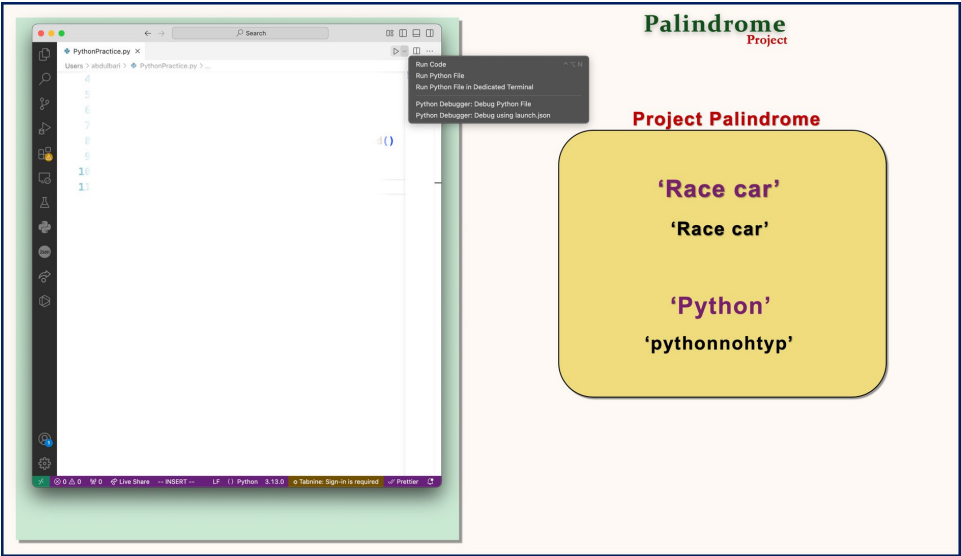
Project Palindrome

'Race car'

'Race car'









PythonPractice.py X

Users > abdulbar1 > PythonPractice.py > ...

1 2 3 4 5 6 7 8 9 10 11

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ... Python Python 3.13.0 Tabnine Sign-in is required Practice

Run Code
Run Python File
Run Python File in Dedicated Terminal
Python Debugger: Debug Python File
Python Debugger: Debug using launch.json

Palindrome
Project

Project Palindrome

'Race car'

'Race car'

'Python'

'pythonnohtyp'

PythonPractice.py X

Users > abdulbar1 > PythonPractice.py > ...

1 2 3 4 5 6 7 8 9 10 11

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ... Python Python 3.13.0 Tabnine Sign-in is required Practice

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Palindrome
Project

Project Palindrome

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'Race car'

'Python'

'pythonnohtyp'

PythonPractice.py X

Users > abdulbari > PythonPractice.py > ...

1 2 3 4 5 6 7 8 9 10 11

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ... Python Python 3.13.0

```
#!/usr/local/bin/python3
#users/abdu/PythonPractice.py
(base) abdulbari@Abduls-MacBook-Pro-2 ~ % /usr/local/bin/python3 /Users/abdu/PythonPractice.py
Enter a Phrase Race car
Race car
(base) abdulbari@Abduls-MacBook-Pro-2 ~ % /usr/local/bin/python3 /Users/abdu/PythonPractice.py
Enter a Phrase Python
pythonnohtyp
(base) abdulbari@Abduls-MacBook-Pro-2 ~ %
```

Palindrome

Project

Project Palindrome

'Race car'

'Race car'

'Python'

'pythonnohtyp'

Palindrome

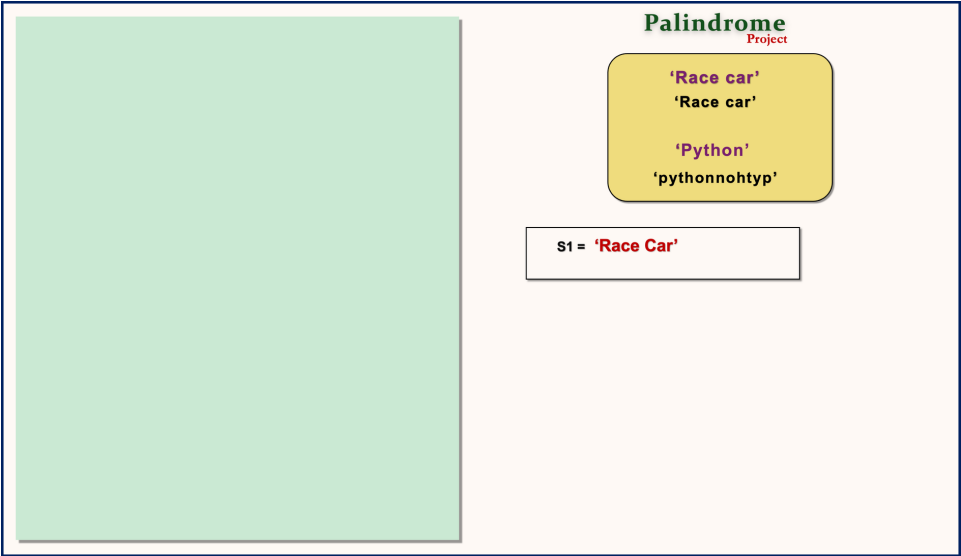
Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'



Palindrome
Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

S1 = 'Race Car'

S2 = S1.replace(' ','')

Palindrome
Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

S1 = 'Race Car'

S2 = S1.replace(' ','')

rev =

Palindrome

Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

S1 = 'Race Car'

S2 = S1.replace(' ','')

rev = S2[::-1]

Palindrome

Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

S1 = 'Race Car'

S2 = S1.replace(' ','')

rev = S2[::-1]

if

Palindrome Project

'Race car'
'Race car'

'Python'
'pythonnohtyp'

```
S1 = 'Race Car'  
S2 = S1.replace(' ','')  
rev = S2[::-1]  
if S2.casefold() == rev:
```

Palindrome Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

```
S1 = 'Race Car'
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Palindrome

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```

Palindrome

Project

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'Race car'

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Palindrome Project

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```

Palindrome Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

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if S2.casefold() == rev.casefold():
    print(S1)
else:
    palindrome = S2.casefold()
```

Palindrome Project

'Race car'

'Race car'

'Python'

'pythonnohtyp'

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rev = S2[::-1]
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    print(S1)
else:
    palindrome = S2.casefold() + rev.casefold()
```

Palindrome

Project

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